One collapse law for every root of C_N: the gauge theorem, and the (Q)-divisors

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Abstract

Let $\psi(v) = (1 - qv^h)/(1 - q)$, $\varphi = \psi^{-1}$, and $\widehat{A}_{\mathbf{M},\mathbf{I}} := \widetilde{A}_{\mathbf{M},\mathbf{I}} \prod_{m \in \mathbf{M}} \psi(w_m)$, where $\widetilde{A}$ is Motegi's $\mathcal{E}^X$-normalised type-X weight function on the 2^N fixed points of $LG(N, 2N)$. I prove the *gauge theorem*: for every root α of C_N the collapse multiplier is $\rho^\alpha = \prod_{m \in D(\alpha)} \psi(w_m)^{-1}$, where $D(\alpha)$ is the set of labels the collision at α deletes — so $\widehat{A}$ collapses with multiplier exactly 1 at every root, for every N. The three root families are not three cases. They are the three orbit types of the *inversion involution* $\iota : v \mapsto v^{-1}$ on $D(\alpha)$: empty $(e_i - e_j)$, a fixed point $(h\, e_m)$, a free 2-orbit $(e_a + e_b)$; and the theorem is the $|D| = 0, 1, 2$ specialisation of one lemma valid for *every* ι-closed multiset. Two reindexing identities along ι do all the work. This reproves (QC) and (SRC) by a single argument. One hypothesis I omitted in a first draft — that the spectral values be pairwise distinct — was refuted by my own plant census and is restored, which also retracts one of the three clauses with which I first answered the pre-registered multiset falsifier; the falsifier still resolves in favour of the set reading, by the other two.

Separately, and this is the result Rick is blocked on: **my own brief was wrong about what the (Q)-divisors are**. They are the divisors $\{1 - q \cdot \text{monomial} = 0\}$ of the diagonal, *not* $\{w_a w_b = 1\}$, which his Corollary 1 already covers and lists. So Task 0 does not answer his question 1. I answer it instead with Theorem Q: at a (Q)-divisor the lattice is cut out by *vanishing* conditions rather than congruences, vanishing conditions are automatically multiplicative, and therefore no GKM edge is needed — his "no GKM edge controls them" is right, and none is required. Combined with the root-hyperplane half this gives, for all N, $\hat{c}^{\nu}_{\lambda\mu} \in \mathbb{C}(q)[w_1^{\pm}, \ldots, w_N^{\pm}]$ — every structure constant of $\widehat{A}$ is a Laurent polynomial — which promotes my computed $0/115$ at $N = 3$ to a theorem. Credit where it is due twice over: the vanishing statements this rests on are Motegi's own divisibility propositions (L1083–1091, L3302–3315), which I found only because Rick's Theorem Ψ cites one of them and the brief told me to check his attributions first-hand — and reading it closed a gap I had filed in this paper's own first draft.

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1 What this session set out to prove, and what happened

The brief set one headline theorem (Task 0, the gauge theorem) and three subordinate tasks. Scoring, before anything else:

- **Task 0 — proved**, and it is *not* the nearly-formal statement the brief predicted it might be. The brief's step 2 guessed that the content is just "$\Psi(\mathbf{M})/\Psi(\mathbf{M} \setminus D) = \prod_{m \in D} \psi(w_m)$, hence nearly formal". That half *is* formal. The content is step 3, and it has a single uniform mechanism (§3–§6) which I did not have when the brief was written.
- **Task 1 — the brief's premise is false, and the question is answered anyway.** The brief asserted that the (Q)-divisors are the pair divisors $\{w_a w_b = 1\}$ and therefore that Task 0 settles question 1. Reading Rick's document first-hand (as the brief instructed) shows "(Q)-factor" denotes a factor $1 - q \cdot$monomial; his Corollary 1 *lists* $\{w_a w_b = 1\}$ among what it covers. See §9. Theorem Q (§10) answers question 1 affirmatively, by a different argument, and Corollary 28 answers it in the strongest available form.

- **Task 2 — derived, not opined** (§11), as a corollary of Theorem Q rather than of Task 0.
- **Task 3 (Q505) — verdict *unusable***, with the domain of each instrument stated and one instrument unavailable by session rule (§13). Unusable is not survival and I do not record one.

Two instrument faults are recorded where they occurred: a radical in a uniformiser made two order-measurements of the *same object* disagree (§15, V6), and the "four copies verified as a set" claim of the previous session was measured over the wrong universe (§14).

2 Setup

I keep the notation of `2026-10-18-atype-divisors-and-the-gkm-lattice.tex` and `2026-10-17-c2-coboundar` including the corrected divisor nomenclature: divisors are named by the *root* that cuts them, never by a root length, because long/short is a function of h and these statements quantify over h.

Let $R := \mathbb{C}(q)[w_1^{\pm}, \dots, w_N^{\pm}]$, $F := \operatorname{Frac} R = \mathbb{C}(q)(w_1, \dots, w_N)$, and $P := 2^{[N]}$, identified with the T-fixed points of $LG(N, 2N)$. Motegi's symmetrisation formula (`2610.15344`, eq. `Motherfunction`, L474–487) is

$$F_{N,n,r}(u \mid w \mid \mathbf{I}) \;=\; \frac{1}{(n-r)!}\Big(\prod_{k=1}^{n} u_k^{-(N+n)}\Big) \sum_{\sigma \in S_n} T\big(u_{\sigma(1)}, \dots, u_{\sigma(n)}\big), \tag{1}$$

where, writing $v_j := u_{\sigma(j)}$ and calling slots $1, \dots, r$ **heads** and $r+1, \dots, n$ **tails**,

$$\begin{aligned} T(v) = {} & \underbrace{\prod_{j=1}^{r}\Big[(1-q)(v_j^2-1)\prod_{k=i_j+1}^{N}(v_j - qw_k)\prod_{k=1}^{i_j-1}(v_j - w_k)\Big]}_{\text{line 1}} \cdot \underbrace{\prod_{1\le j<k\le r}\frac{qv_j - v_k}{v_j - v_k}(v_jv_k - 1)}_{\text{line 2a}} \\ & \cdot \underbrace{\prod_{j>r}\prod_{k=1}^{N}(v_j - qw_k)}_{\text{line 2b}} \cdot \underbrace{\prod_{j>r}\prod_{k\le r}\frac{qv_j - v_k}{v_j - v_k}(v_jv_k - 1)}_{\text{line 3a}} \cdot \underbrace{\prod_{r<j<k\le n}(v_jv_k - q)}_{\text{line 3b}} \cdot \underbrace{\prod_{j>r}(\beta v_j - \gamma)}_{\text{line 3c}}. \end{aligned} \tag{2}$$

Type C has $\beta = 1$, $\gamma = 0$, $h = 2$ and normaliser (eq. `typeCnormalizationfactor`, L1057–1062)

$$\mathcal{E}^{C}_{N,n,r}(u, w) = \frac{\overbrace{\prod_{j,k=1}^{n}\big(1 - q\tfrac{u_k}{u_j}\big)}^{Q_1} \cdot \prod_{j=1}^{n}\prod_{k=1}^{N}(1 - qu_jw_k)}{\underbrace{\prod_{1\le j<k\le n}(1 - qu_ju_k)}_{Q_2} \cdot \prod_{1\le j\le k\le N}(1 - qw_jw_k) \cdot (1-q)^{\,n-r}}, \tag{3}$$

and $\widetilde{A} = W^C/\mathcal{E}^C$ with $W^C = F_{N,n,r}$. Everything in §3–§6 is type C; type B is §12.

Two facts from Motegi, both deep-read:

(M3) Triangularity. $\widetilde{A}_{\mathbf{M},\mathbf{I}} \neq 0$ only if $\mathbf{M} \succeq \mathbf{I}$ in Motegi's order, **defined at L582–589** as $|\mathbf{M}| \ge |\mathbf{I}|$ and $m_{s-r+j} \ge i_j$ for all j; the vanishing is proved at L670–675, and $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ when $|\mathbf{M}| < |\mathbf{I}|$ (L2843).

(M4) The diagonal. $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is the product over positive roots (eq. `normalizationWC`, L1078), written out at L997–1030:

$$\widetilde{A}_{\mathbf{I},\mathbf{I}} = \prod_{1\le a\le b\le r}\Big(1-\tfrac{1}{w_{i_a}w_{i_b}}\Big)\cdot\prod_{\substack{c\in\mathbf{I}^c\\ i_l>c}}\Big(1-\tfrac{w_c}{w_{i_l}}\Big)\cdot\prod_{\substack{c\le d\\ c,d\in\mathbf{I}^c}}(1-qw_cw_d)\cdot\prod_{\substack{c\in\mathbf{I}^c\\ i_l<c}}\Big(1-q\tfrac{w_c}{w_{i_l}}\Big). \tag{4}$$

By (M3)+(M4) the matrix $\Phi=(\widetilde{A}_{\mathbf{M},\mathbf{I}})$ is triangular with nonzero diagonal, so $\{\phi_{\mathbf{I}}\}$, $\phi_{\mathbf{I}}(\mathbf{M}):=\widetilde{A}_{\mathbf{M},\mathbf{I}}$, is an F-basis of F^P, $\det\Phi=\pm\prod_{\mathbf{I}}\widetilde{A}_{\mathbf{I},\mathbf{I}}$, and the structure constants $c^{\nu}_{\lambda\mu}$ are defined by $\phi_\lambda\phi_\mu=\sum_\nu c^{\nu}_{\lambda\mu}\phi_\nu$.

Remark 1 (the two divisor classes of (4), named once and for all)**.** The irreducible factors of (4) fall into exactly two classes, and every statement below is indexed by which one it is about.

Root hyperplanes. $\{w_aw_b=1\}$ $(a<b$; root $e_a+e_b)$, $\{w_a^2=1\}$ $(a=b$; root $2e_a)$, $\{w_c=w_{i_l}\}$ (root $e_c-e_{i_l}$). These are the q-free factors. They are $W(C_N)$-stable and each is the vanishing locus of a T-curve character.

(Q)-divisors. $\{1-qw_cw_d=0\}$ $(c<d)$, $\{1-qw_c^2=0\}$, $\{1-qw_c/w_{i_l}=0\}$. These are the factors carrying a q. They are *not* root hyperplanes, not $W(C_N)$-stable, and no T-curve cuts them. In the affine picture they are the walls $\alpha+\delta=0$ with $q=e^{-\delta}$.

This is the distinction my brief collapsed, and §9 records how. Ψ adds no new class: its divisors $\{1-qw_m^2=0\}$ are already the second (Q)-family.

3 The deletion map, and the inversion involution

Definition 2 (the collision data of a root)**.** For a positive root α of C_N let $H_\alpha\subset\operatorname{Spec}R$ be the root hyperplane it cuts, s_α its action on P, and $D(\alpha)\subseteq[N]$ the set of labels that s_α *deletes*:

root	H_α	$s_\alpha(\mathbf{M})$ on the colliding stratum	$D(\alpha)$
e_i-e_j	$\{w_i=w_j\}$	$\mathbf{M}\triangle\{i,j\}$ $(\lvert\mathbf{M}\cap\{i,j\}\rvert=1)$	$\varnothing$
$h\,e_m$	$\{w_m^h=1\}$	$\mathbf{M}\smallsetminus\{m\}$ $(m\in\mathbf{M})$	$\{m\}$
e_a+e_b	$\{w_aw_b=1\}$	$\mathbf{M}\smallsetminus\{a,b\}$ $(\{a,b\}\subseteq\mathbf{M})$	$\{a,b\}$

Lemma 3 (the model is the signed-permutation action, and is consistent with (QC) and (SRC))**.** *Identify $\mathbf{M}\in P$ with the isotropic subspace spanned by e_m $(m\in\mathbf{M})$ and f_m $(m\notin\mathbf{M})$, so that $W(C_N)$ acts on P through signed permutations. Then $s_{h\,e_m}$ is the sign change at m, $s_{e_i-e_j}$ is the transposition, and $s_{e_a+e_b}=$ (sign change at a)(sign change at b)$(a\,b)$ acts by*

$$\mathbf{M}\mapsto\begin{cases}\mathbf{M}\smallsetminus\{a,b\} & \{a,b\}\subseteq\mathbf{M},\\ \mathbf{M}\cup\{a,b\} & \mathbf{M}\cap\{a,b\}=\varnothing,\\ \mathbf{M} & \textit{otherwise.}\end{cases}$$

In particular Definition 2 is exactly the pairing used in (QC) *and* (SRC) *of `2026-10-17-c2`, so what was a Remark there is now a lemma.*

Proof. Only the third case needs checking. If $\{a,b\}\subseteq\mathbf{M}$ the transposition fixes $\mathbf{M}$ and the two sign changes delete a and b; if $\mathbf{M}\cap\{a,b\}=\varnothing$ they adjoin both; if $a\in\mathbf{M}\not\ni b$ the transposition gives $(\mathbf{M}\smallsetminus\{a\})\cup\{b\}$ and the sign changes restore a and remove b, returning $\mathbf{M}$. The fixed-point/curve dictionary then assigns to each pair $(\mathbf{M},s_\alpha\mathbf{M})$ the T-curve of weight α, i.e. the character w_i/w_j, w_m^h, w_aw_b respectively; H_α is where that character is 1. □

Now the observation the whole paper rests on. Let $\iota : F^\times \to F^\times$, $\iota(v) = v^{-1}$.

Definition 4 (ι-closed)**.** A finite multiset $\mathbf{D}$ of elements of $F^\times$ is **ι-closed** if there is an involution τ of its index set with $y_{\tau(j)} = y_j^{-1}$ for all j. Call τ a *realising involution*; τ-fixed points are the y with $y^2 = 1$.

Lemma 5 (the three root families are the three ι-orbit types)**.** *Let α be a positive root of C_N (type C, $h = 2$), $\mathbf{M}$ in its colliding stratum, and $\mathbf{D}(\alpha) := \{\, w_m : m \in D(\alpha) \,\}$ as a multiset of values, restricted to H_α. Then $\mathbf{D}(\alpha)$ is ι-closed, and*

$$\alpha = e_i - e_j \;\Rightarrow\; \mathbf{D} = \varnothing; \qquad \alpha = 2e_m \;\Rightarrow\; \mathbf{D} = \{\varepsilon\},\ \varepsilon^2 = 1 \text{ (a } \tau\text{-fixed point)}; \qquad \alpha = e_a + e_b \;\Rightarrow\; \mathbf{D} = \{x, x^{-1}\} \text{ (a fr}$$

Moreover the value multiset of $\mathbf{M}$ and of $s_\alpha\mathbf{M}$ satisfy, on H_α, $w_{\mathbf{M}} = w_{s_\alpha\mathbf{M}} \sqcup \mathbf{D}(\alpha)$.

Proof. On $\{w_m^2 = 1\}$, $\varepsilon := w_m$ satisfies $\varepsilon^{-1} = \varepsilon$; on $\{w_a w_b = 1\}$, $w_b = w_a^{-1}$; and $\varnothing$ is ι-closed. For the last statement: in the second and third cases $s_\alpha\mathbf{M} = \mathbf{M} \smallsetminus D(\alpha)$ by Definition 2, so the value multisets differ by exactly $\mathbf{D}(\alpha)$. In the first case $|\mathbf{M} \cap \{i, j\}| = 1$, say $i \in \mathbf{M} \not\ni j$; then $s_{ij}\mathbf{M} = (\mathbf{M} \smallsetminus \{i\}) \cup \{j\}$ and on $\{w_i = w_j\}$ the two value multisets are *equal*, which is the $\mathbf{D} = \varnothing$ statement. □

Remark 6 (what this replaces)**.** The previous paper's Remark (`rem:mech`) observed that only the A-type collision preserves arity, concluded "arity preserved $\Rightarrow \rho = 1$ by symmetry; arity changed $\Rightarrow$ go and compute", and recorded that the two computations landed differently. Lemma 5 says the arity drop is not merely a discriminator: it is $|\mathbf{D}(\alpha)|$, the *size of an ι-orbit*, and ι-closedness — not arity — is the hypothesis that makes the computation go. The boundary family, which was the counterexample to the tempting strengthening, is the τ-fixed-point orbit; there is no third kind of orbit, which is why there is no fourth root family.

4 Head exclusion

Lemma 7 (head exclusion)**.** *Let $\mathbf{D}$ be ι-closed with realising involution τ, let $\mathbf{M}$ have value multiset $\mathbf{R} \sqcup \mathbf{D}$ **with pairwise distinct values**, and fix $\mathbf{I}$ with $r = |\mathbf{I}|$. In $\sum_{\sigma \in S_n} T(u_\sigma)$ of (1), every σ that places some slot of $\mathbf{D}$ in a head position contributes 0.*

Proof. Collect the factors of (2) of the shape $(\cdot - 1)$. Line 1 contributes $(v_j^2 - 1)$ for each head j; line 2a contributes $(v_j v_k - 1)$ for each pair of heads $j < k$; line 3a contributes $(v_j v_k - 1)$ for each pair (tail, head). Hence:

> *the factor $(u_p u_{p'} - 1)$ occurs in $T(u_\sigma)$ for every unordered pair of slots $\{p, p'\}$ — possibly $p = p'$ — at least one of which is a head.*

(The remaining pairs, tail–tail, carry $(v_j v_k - q)$ from line 3b instead.) Distinctness is used exactly here: the slot pair $\{p, p'\}$ also carries the *denominator* $(v_p - v_{p'})$ from line 2a or 3a, and distinctness is the statement that it does not vanish. See Remark 8.

Now let slot p of $\mathbf{D}$ lie in a head position. If $\tau(p) = p$ then $u_p^2 = 1$ and the pair $\{p, p\}$ gives $(u_p^2 - 1) = 0$. If $\tau(p) \neq p$ then $u_p u_{\tau(p)} = 1$ and the pair $\{p, \tau(p)\}$ — which contains the head p, whether or not $\tau(p)$ is a head — gives $(u_p u_{\tau(p)} - 1) = 0$. Either way $T(u_\sigma) = 0$. □

Remark 8 (the hypothesis I omitted in the first draft, and what found it)**.** My first draft of Lemma 7 and Theorem 14 had no distinctness hypothesis, and the fast census (§15, V2) refuted

it. The structure of the refutation is sharp and worth stating, because the omitted hypothesis is about the *same factor* the lemma uses.

Head exclusion works by exhibiting a vanishing *numerator* factor $(v_p v_{p'} - 1)$ for a slot pair $\{p, p'\}$ with p a head. But lines 2a and 3a attach to that very pair the *denominator* $(v_p - v_{p'})$. If $v_p = v_{p'}$ — which for ι-closed $\mathbf{D}$ happens exactly when two values of $\mathbf{D}$ coincide, i.e. when $v_p = v_{p'} = \varepsilon$ with $\varepsilon^2 = 1$ — then numerator *and* denominator vanish and the term is $0/0$, not 0. So the lemma's conclusion fails not because the mathematics is different but because the expression is undefined.

Measured, and the pattern is exact: with $\mathbf{D} = \{\varepsilon, \varepsilon\}$ the adjunction law is green at $r = 0$ in 12/12 instances and **fails in** 12/12 **at** $r = 1$ **and** 2/2 **at** $r = 2$; with $\mathbf{D} = \{x, x^{-1}, y, y^{-1}\}$ (four distinct values, $d = 4$) it is green at every r, 7/7. $r = 0$ is exactly the case with no head slot, hence with no line-2a/3a denominator at all. **So the discriminator is a head slot, which is the discriminator the lemma itself names** — a hypothesis and its failure mode produced by one factor.

Remark 9 (this is one step, not two)**.** Step 1 of Rick's Lemma QC ("both colliding indices must be tails") and Step 1 of my Theorem (SRC) ("ε must be a tail") are the two cases $\tau(p) \neq p$ and $\tau(p) = p$ of Lemma 7. They were proved separately, by looking at different factors: the $(v_j v_k - 1)$ of line 2a/3a in one case, the $(v_j^2 - 1)$ of line 1 in the other. They are the *same* factor family $\prod_{1 \le j \le k \le r} (v_j v_k - 1)$ — the positive roots of C_r in the head variables, with $j = k$ the long/short diagonal term — and the collision locus of α is the vanishing locus of exactly the one factor attached to α.

Corollary 10 (the head selections match)**.** *T is symmetric under permuting the tail slots, so with $d := |\mathbf{D}|$, $s := |\mathbf{R}| \geq r$,*

$$\frac{1}{(n-r)!} \sum_{\sigma \in S_n} T(u_\sigma) \;=\; \sum_{\substack{(p_1,\ldots,p_r)\ \text{distinct} \\ \text{slots of } \mathbf{R}}} T\big(p_1, \ldots, p_r;\ \textit{tails} = \textit{the rest}\big),$$

and the index set on the right is the same *for $\mathbf{M}$ and for $\mathbf{M} \smallsetminus D$. (If $s < r$ both sides vanish: $\widetilde{A}_{\mathbf{M},\mathbf{I}} = \widetilde{A}_{\mathbf{R},\mathbf{I}} = 0$ by* (M3)*, and the statement below is the vacuous $0 = \lambda \cdot 0$.)*

Proof. Lines 2b, 3b, 3c are symmetric in the tails and line 3a is a product over tails of head-dependent factors, so T is tail-symmetric and the $n - r$ tail orderings give $(n - r)!$ equal terms. By Lemma 7 the surviving head selections avoid $\mathbf{D}$ entirely, i.e. are head selections from $\mathbf{R}$. □

5 Two reindexing identities

Fix $\mathbf{D}$ ι-closed, $d = |\mathbf{D}|$, realising involution τ, and set

$$Q_1(\mathbf{D}) := \prod_{\text{ordered } (j,k)} \Big(1 - q\tfrac{y_k}{y_j}\Big), \quad Q_2(\mathbf{D}) := \prod_{\{j,k\},\, j \neq k} (1 - q y_j y_k), \quad P_2(\mathbf{D}) := \prod_{\{j,k\},\, j \neq k} (y_j y_k - q),$$

the products over all ordered pairs of slots of $\mathbf{D}$ including $j = k$ (for Q_1), resp. all unordered pairs of distinct slots.

Lemma 11. $P_2(\mathbf{D}) = \big(\prod_{y \in \mathbf{D}} y\big)^{d-1} Q_2(\mathbf{D})$.

Proof. $y_jy_k - q = y_jy_k\big(1 - q(y_jy_k)^{-1}\big)$, and $\prod_{\{j,k\},j\neq k} y_jy_k = \big(\prod_y y\big)^{d-1}$ since each slot occurs in $d-1$ pairs. The map $\{j,k\} \mapsto \{\tau j, \tau k\}$ is an involution of the set of unordered pairs of distinct slots, and $(y_{\tau j}y_{\tau k})^{-1} = y_jy_k$; so $\prod_{\{j,k\}}\big(1 - q(y_jy_k)^{-1}\big) = \prod_{\{j,k\}}(1 - qy_jy_k) = Q_2(\mathbf{D})$. □

Lemma 12. $\dfrac{Q_2(\mathbf{D})^2}{Q_1(\mathbf{D})} = \dfrac{1}{\prod_{y\in\mathbf{D}}(1-qy^2)}$.

Proof. Split off the diagonal of Q_1: $Q_1 = (1-q)^d \prod_{(j,k):j\neq k}(1 - qy_k/y_j)$. Since $y_j^{-1} = y_{\tau j}$ we have $y_k/y_j = y_ky_{\tau j}$, so substituting $j = \tau(i)$ (a bijection) turns the product over $\{(j,k) : j \neq k\}$ into $\prod_{(i,k):\,\tau i\neq k}(1-qy_iy_k)$. Meanwhile $Q_2^2 = \prod_{(j,k):\,j\neq k}(1-qy_jy_k)$ over *ordered* pairs. The two products have the same ambient index set and differ only in which diagonal-type pairs are excluded, so

$$\frac{Q_2^2}{\prod_{(i,k):\tau i\neq k}(1-qy_iy_k)} = \frac{\prod_{k=\tau j}(1-qy_jy_k)}{\prod_{k=j}(1-qy_jy_k)} = \frac{\prod_j(1-q\,y_jy_{\tau j})}{\prod_j(1-qy_j^2)} = \frac{(1-q)^d}{\prod_j(1-qy_j^2)},$$

using $y_jy_{\tau j} = 1$. Hence $Q_2^2/Q_1 = (1-q)^{-d}\cdot(1-q)^d/\prod_j(1-qy_j^2)$. □

Remark 13. Lemma 12 is where ψ comes from. Its right-hand side is $\prod_{y\in\mathbf{D}}(1-qy^2)^{-1}$: a product, over the deleted values, of a function of *one* variable. Nothing in the derivation chose that shape; it is forced by the fact that τ pairs the Q_1-denominators with the Q_2-numerators. The coboundary property of ρ (Theorem A of `2026-10-17-c2`) is a consequence, not an observation.

6 The gauge theorem

Theorem 14 (ι-adjunction; the gauge theorem)**.** *Let* $\mathbf{R}$ *be a finite multiset of values with* $|\mathbf{R}| \geq r = |\mathbf{I}|$*, let* $\mathbf{D}$ *be* ι*-closed with* $d = |\mathbf{D}|$*,* ***assume the values of*** $\mathbf{R}\sqcup\mathbf{D}$ ***are pairwise distinct*** *(Remark 8), and let* w *be the alphabet, at a generic point of the locus on which* $\mathbf{D}$ *is* ι*-closed. Write* $\widetilde{A}(\mathbf{U}) := W^C_{N,|\mathbf{U}|,r}(\mathbf{U} \mid w \mid \mathbf{I})/\mathcal{E}^C_{N,|\mathbf{U}|,r}(\mathbf{U}, w)$*. Then*

$$\boxed{\widetilde{A}(\mathbf{R}\sqcup\mathbf{D}) \;=\; \Big(\prod_{y\in\mathbf{D}}\varphi(y)\Big)\widetilde{A}(\mathbf{R}), \qquad \varphi(y) = \frac{1-q}{1-qy^2}}$$

Proof. Fix a head selection $\mathbf{h} \subset \mathbf{R}$, $|\mathbf{h}| = r$, and put $\mathbf{R}' := \mathbf{R} \smallsetminus \mathbf{h}$. By Corollary 10 the head selections are the same on both sides; I compare the two summands term by term.

Step 1 (the T-ratio). Lines 1 and 2a depend only on the heads and w, so they cancel. The extra factors carried by the d tails of $\mathbf{D}$ are: line 2b, $\prod_{y\in\mathbf{D}}\prod_{k=1}^N(y-qw_k)$; line 3a, $\prod_{y\in\mathbf{D}}A(y)$ with $A(y) := \prod_{v\in\mathbf{h}}\frac{qy-v}{y-v}(yv-1)$; line 3b, $\prod_{y\in\mathbf{D}}\prod_{t\in\mathbf{R}'}(yt-q)\cdot P_2(\mathbf{D})$; line 3c, $\prod_{y\in\mathbf{D}} y$ (as $\beta = 1, \gamma = 0$).

Step 2 (the head-dependence cancels, by ι). Using $yv - 1 = y(v - y^{-1})$,

$$\prod_{y\in\mathbf{D}}A(y) = \Big(\prod_y y^r\Big)\prod_{v\in\mathbf{h}}\prod_{y\in\mathbf{D}}\frac{(qy-v)(v-y^{-1})}{y-v}.$$

Because $\mathbf{D}$ is ι-closed, $\prod_{y\in\mathbf{D}}(v-y^{-1}) = \prod_{y\in\mathbf{D}}(v-y)$ (reindex by τ), while $\prod_{y\in\mathbf{D}}(y-v) = (-1)^d\prod_{y\in\mathbf{D}}(v-y)$; so for each v the ratio $\prod_y(v-y^{-1})/(y-v)$ equals $(-1)^d$. Together with $\prod_y(qy-v) = (-1)^d\prod_y(v-qy)$ this gives

$$\prod_{y\in\mathbf{D}}A(y) = \Big(\prod_{y\in\mathbf{D}}y^r\Big)\prod_{y\in\mathbf{D}}\prod_{v\in\mathbf{h}}(v-qy). \tag{5}$$

The same reindexing on line 3b gives $\prod_y \prod_{t\in\mathbf{R}'}(yt-q) = \big(\prod_y y^{\,s-r}\big)\prod_y \prod_{t\in\mathbf{R}'}(t-qy)$. Multiplying (5) by this **merges the head product with the tail product into a product over all of R**:

$$\prod_{y\in\mathbf{D}} A(y)\cdot\prod_{y\in\mathbf{D}}\prod_{t\in\mathbf{R}'}(yt-q) = \Big(\prod_y y^{\,s}\Big)\prod_{y\in\mathbf{D}}\prod_{v\in\mathbf{R}}(v-qy),$$

which is independent of $\mathbf{h}$. Hence so is the whole T-ratio, and it factors out of the sum over head selections.

Step 3 (the prefactor, and F). With $n = s+d$, the prefactor ratio is $\prod_{y\in\mathbf{D}} y^{-(N+n)}\cdot\prod_{v\in\mathbf{R}} v^{-d}$. Collecting, and using $v^{-1}(v-qy) = 1-qy/v$, $y^{-N}\prod_k(y-qw_k) = \prod_k(1-qw_k/y)$ and once more the τ-reindexing $\prod_y\prod_k(1-qw_k y^{-1}) = \prod_y\prod_k(1-qw_k y)$:

$$\frac{F_{N,n,r}(\mathbf{R}\sqcup\mathbf{D})}{F_{N,s,r}(\mathbf{R})} = \Big(\prod_{y\in\mathbf{D}} y\Big)^{1-d} P_2(\mathbf{D})\prod_{y\in\mathbf{D}}\Big[\prod_{v\in\mathbf{R}}\big(1-\tfrac{qy}{v}\big)\prod_{k=1}^{N}(1-qyw_k)\Big]. \tag{6}$$

Step 4 (the normaliser). From (3), the factors of $\mathcal{E}^C$ acquired on adjoining $\mathbf{D}$ are: in the numerator, $Q_1(\mathbf{D})\cdot\prod_y\prod_{v\in\mathbf{R}}\big(1-q\tfrac{v}{y}\big)\cdot\prod_y\prod_{v\in\mathbf{R}}\big(1-q\tfrac{y}{v}\big)\cdot\prod_y\prod_k(1-qyw_k)$; in the denominator, $Q_2(\mathbf{D})\cdot\prod_y\prod_{v\in\mathbf{R}}(1-qyv)\cdot(1-q)^d$. By τ-reindexing, $\prod_y\prod_v(1-qv/y) = \prod_y\prod_v(1-qvy)$, which cancels the denominator's $\prod_y\prod_v(1-qyv)$. Therefore

$$\frac{\mathcal{E}^C_n}{\mathcal{E}^C_s} = \frac{Q_1(\mathbf{D})}{Q_2(\mathbf{D})\,(1-q)^d}\prod_{y\in\mathbf{D}}\Big[\prod_{v\in\mathbf{R}}\big(1-\tfrac{qy}{v}\big)\prod_{k=1}^{N}(1-qyw_k)\Big]. \tag{7}$$

Step 5. The bracketed products in (6) and (7) are identical and cancel, leaving

$$\frac{\widetilde{A}(\mathbf{R}\sqcup\mathbf{D})}{\widetilde{A}(\mathbf{R})} = \Big(\prod_y y\Big)^{1-d}(1-q)^d\,\frac{P_2(\mathbf{D})\,Q_2(\mathbf{D})}{Q_1(\mathbf{D})} \overset{\text{Lem.11}}{=} (1-q)^d\frac{Q_2(\mathbf{D})^2}{Q_1(\mathbf{D})} \overset{\text{Lem.12}}{=} \prod_{y\in\mathbf{D}}\frac{1-q}{1-qy^2}. \qquad \square$$

Corollary 15 (ρ^α for every root, and $\widehat{A}$'s multiplier)**.** *For every N, every positive root α of C_N, every $\mathbf{M}$ in the colliding stratum and every $\mathbf{I}$, on H_α*

$$\widetilde{A}_{\mathbf{M},\mathbf{I}} = \rho^\alpha\,\widetilde{A}_{s_\alpha\mathbf{M},\mathbf{I}}, \qquad \rho^\alpha = \prod_{m\in D(\alpha)}\varphi(w_m) = \prod_{m\in D(\alpha)}\psi(w_m)^{-1},$$

and consequently $\widehat{A}_{\mathbf{M},\mathbf{I}} = \widehat{A}_{s_\alpha\mathbf{M},\mathbf{I}}$ on H_α: the collapse multiplier of $\widehat{A}$ is exactly 1 at every root of C_N, for all N.

Proof. Lemma 5 gives $w_{\mathbf{M}} = w_{s_\alpha\mathbf{M}}\sqcup\mathbf{D}(\alpha)$ with $\mathbf{D}(\alpha)$ ι-closed. Theorem 14's distinctness hypothesis holds at the generic point of H_α: $|\mathbf{D}(\alpha)|\le 2$, and when $|\mathbf{D}| = 2$ its values satisfy $w_a w_b = 1$, so they coincide only on $\{w_a^2 = 1\}\cap H_{ab}$, a proper closed subvariety of H_{ab}; the values of $w_{s_\alpha\mathbf{M}}$ are distinct indeterminates. Theorem 14 with $\mathbf{R} = w_{s_\alpha\mathbf{M}}$ then gives the multiplier, and both sides lie in $\mathcal{O}_{H_\alpha}$, so the identity at the generic point is the identity in the local ring. For $\widehat{A}$: $\Psi(\mathbf{M}) = \Psi(s_\alpha\mathbf{M})\prod_{m\in D(\alpha)}\psi(w_m)$ because $\mathbf{M}$ and $s_\alpha\mathbf{M}$ differ exactly in $D(\alpha)$, so the two factors are mutually inverse. (For $\alpha = e_i - e_j$, $D = \varnothing$ and $\Psi(\mathbf{M}) = \Psi(s_{ij}\mathbf{M})$ on $\{w_i = w_j\}$ because $\psi(w_i) = \psi(w_j)$ there.) $\square$

Corollary 16 (the three known cases, recovered)**.** *Specialising Corollary 15:*

α	$\lvert D\rvert$	$\mathbf{D}$ *on* H_α	ρ^α	*prior statement*
$e_i - e_j$	0	$\varnothing$	1	*Lemma A1 (u-symmetry)*
$2e_m$	1	$\{\varepsilon\}, \varepsilon^2 = 1$	$\varphi(\varepsilon) = \frac{1-q}{1-q} = 1$	*Theorem (SRC)*
$e_a + e_b$	2	$\{x, x^{-1}\}$	$\frac{(1-q)^2}{(1-qx^2)(1-qx^{-2})} = g$	*(QC), Rick's Lemma QC*

Remark 17 (the $d = 0$ case *is* u-symmetry)**.** At $\mathbf{D} = \varnothing$ Theorem 14 reads $\widetilde{A}(\mathbf{R}) = \widetilde{A}(\mathbf{R})$, which is vacuous as an identity — the content in the A-type case sits entirely in the second clause of Lemma 5, that the two value multisets coincide on H_α. That clause is u-symmetry. So the correct statement of the unification is: $\widetilde{A}_{\cdot,\mathbf{I}}$ *is a function of the value multiset (symmetry), and adjoining an* ι*-closed multiset multiplies it by* $\prod\varphi$ *(Theorem 14)*; the first half carries the A-type family and the second the other two. I say this plainly because the tempting overstatement — "all three are instances of one identity" — is false at $d = 0$.

7 Three consequences, including the pre-registered falsifier

Corollary 18 (the collision cochain is a coboundary, multiplicatively)**.** $\mathbf{D} \mapsto \prod_{y\in\mathbf{D}} \varphi(y)$ *is multiplicative in* $\mathbf{D}$*:* $\Lambda(\mathbf{D}\sqcup\mathbf{D}') = \Lambda(\mathbf{D})\Lambda(\mathbf{D}')$ *whenever both are* ι*-closed. Hence the collapse multipliers are consistent along any composite of collisions, and* ρ *is the coboundary of* $\Psi^{-1} = \prod\varphi$ *on the collision groupoid.*

Corollary 19 (the falsifier, resolved — and clause (3) of my first draft retracted)**.** *The brief pre-registered: "*$2e_m$ *and* $e_a + e_b$ *share a label when* $a = m$*; go to a point of* $\{w_a^2 = 1\} \cap \{w_a w_b = 1\}$*; if the right object is a multiset rather than a set, the formula is wrong as stated." The set reading is correct, for two reasons, and a third reason I wrote first is* ***wrong****.*

1. *At such a point* $w_a = w_b = \varepsilon$*,* $\varepsilon^2 = 1$*. The formula* $\rho^\alpha = \prod_{m\in D(\alpha)} \varphi(w_m)$ *is a product over the* label set $D(\alpha) = \{a, b\}$*. Labels remain distinct when their values collide, so multiplicity of* values *never enters the formula.*

2. *The formula is* regular *at the special point:* $\varphi(x)\varphi(x^{-1}) = (1-q)^2/((1-qx^2)(1-qx^{-2}))$ *has value* 1 *at* $x = \varepsilon$*, equal to* $\varphi(\varepsilon)^2 = 1$*. So the pair multiplier at the special point is the limit of the generic one, and Corollary 15 — proved at the generic point of* H_{ab}*, where both sides are regular — already determines it there.*

3. **(RETRACTED.)** *I first wrote: "Theorem 14 is proved for an arbitrary* ι*-closed* multiset*, so* $\mathbf{D} = \{\varepsilon, \varepsilon\}$ *is inside its hypothesis." It is not. By Remark 8,* $\mathbf{D} = \{\varepsilon, \varepsilon\}$ *violates distinctness, Motegi's sum is not even defined there for* $r \geq 1$ *(individual* σ*-terms are* $0/0$*), and the census fails* $12/12$ *at* $r = 1$*. The claim was refuted by the plant census of §15 V2 after it was written down, which is the only reason it is not in the final statement.*

Vacuity gate, discharged honestly. *The brief required me to check, before measuring, that the intersection is nonempty and not inside a locus Lemma A1 already covers. Nonempty: yes,* $w_a = w_b = \varepsilon$ *with all other* w *free. Not inside an A-type locus:* ***no — it fails this half.*** *At such a point* $w_a = w_b$*, so the point lies on the A-type divisor* $\{w_a = w_b\}$ *as well. So the geometric form of the falsifier was always going to be answerable from Lemma A1, and the falsifier as posed had less discriminating power than the brief credited it with. What carries the answer is clause (1) — a statement about the indexing of the formula, not about the geometry — and clause (2)'s regularity. I record the failed gate rather than report a pass.*

Corollary 20 (strictly more than the three cases)**.** *Theorem 14 holds for ι-closed multisets of any size, e.g.* $\mathbf{D} = \{x, x^{-1}, y, y^{-1}\}$ *with* $\Lambda = \varphi(x)\varphi(x^{-1})\varphi(y)\varphi(y^{-1})$*, four distinct values — verified 7/7 at every r (§15, V2). No root of C_N produces $|\mathbf{D}| \geq 3$, so this generality is not used; it is reported because it is what makes the proof uniform, and because it is the part a plant can attack.*

8 Honest accounting: what is new here

- **Not new.** $\rho^C = (1-q)^2/((1-qw_a^h)(1-qw_b^h))$ is Rick's Lemma QC and my `2026-10-16-c2` Thm `collapse`; he had already identified his g with my ρ^C (UID 819) and already observed $g\,\psi(x)\psi(1/x) = 1$ and $\psi(\pm 1) = 1$. His own summary of his Lemma QC is "the whole content is: $\psi(x)\psi(1/x)$ is exactly $1/g$, and $\psi(\pm 1) = 1$". The *values* of the three multipliers are his and mine from earlier sessions, and the formula $\rho^\alpha = \prod_{D(\alpha)} \psi^{-1}$ is a repackaging of them.

- **New.** (i) Lemma 7: the two Step-1 arguments are one step, over the single factor family $\prod_{j\leq k\leq r}(v_j v_k - 1)$; (ii) Lemmas 11–12: the τ-reindexing identities, which is where φ's one-variable product shape comes from; (iii) Theorem 14, which proves (QC) and (SRC) by *one* computation and extends to arbitrary ι-closed multisets — so it is a strict generalisation, not only a reorganisation; (iv) Lemma 5: the three root families are the three ι-orbit types, which is why there are three and no more.

- **The answer to his question "where does ψ come from on Motegi's side?"** From Lemma 12. $\psi(y)^{-1} = \varphi(y)$ is $(1-q)/(1-qy^2)$ because $Q_2^2/Q_1 = \prod_y (1-qy^2)^{-1}$, and the identity holds because the realising involution τ matches the $\prod_{j,k}(1-qu_k/u_j)$ of $\mathcal{E}^C$ against its $\prod_{j<k}(1-qu_ju_k)$. ψ is a ratio of two blocks of the normaliser, read along ι. That is better than "it looks like a boundary K-matrix scalar" only because it is proved; whether it *is* such a scalar is still open and I do not claim it.

- **Non-uniqueness.** He is right that ψ is not unique ($\psi\chi$ with $\chi(x)\chi(1/x) = 1$, $\chi(\pm 1) = 1$ works), and Theorem C of `2026-10-17-c2` plus Corollary `cor:inv` of `2026-10-18-atype` say (COB)+(SR) is the complete list of conditions. Lemma 12 does not single ψ out either — it produces φ as *the* value of the adjunction scalar, which pins ρ, not the gauge.

9 Task 1: the (Q)-divisors, and a false premise in my own brief

Remark 21 (the correction, stated before the mathematics)**.** My brief wrote:

> "The (Q)-factors $(1 - 1/(w_{i_a}w_{i_b}))$ cut $\{w_a w_b = 1\}$ — which is the hyperplane of the root $e_a + e_b$. So Task 0's $|D| = 2$ case **is** the (Q)-divisor case, and if Task 0 lands, this question is answered in the affirmative with a proof."

This is false. Reading Rick's document (`2026-10-18-theorem-psi.pdf`, local copy `peers/rick/proofs/`) first-hand:

> "Corollary 1 … The structure constants $\hat{c}^{\nu}_{\lambda\mu}$ of $\widehat{A}$ are regular on every root hyperplane of C_N: $\{w_a = \pm 1\}$, $\{w_a w_b = 1\}$, $\{w_a = w_b\}$. **Their only possible non-monomial poles lie on $\{1 - q \cdot \text{monomial} = 0\}$: the (Q)-factors of the diagonal, and the $1 - qw_i^2$ of Ψ.**"

So "(Q)-factor" denotes a factor $1 - q \cdot \text{monomial}$ — the q in (Q) is the q — and $\{w_a w_b = 1\}$ is explicitly on the *covered* side of his dichotomy. Consequences:

1. Task 0 does **not** answer his question 1. The $|D| = 2$ case is the pair divisor, which his Corollary 1 already covers.

2. The "tension" the brief instructed me to resolve — his Corollary claiming what his §5 says is open — **does not exist**. Both readings the brief offered ("his count does cover them"; "the corollary is overstated") presuppose the brief's misidentification. His Corollary 1 and his §5 are consistent and both correct.

3. His "no GKM edge controls them" is **true**, and is the right diagnosis: a (Q)-divisor is not $W(C_N)$-stable and is not the vanishing locus of a T-curve character, so there is no edge to use. What §10 shows is that none is needed.

I read the colength step first-hand as instructed, and it is fine: "the D-edges form a partial matching, so the colength of L_D is $\#(D\text{-edges})$" with the owner locator unpinned and treated as standard. I have no objection to it; it is the same count as my `prop:lattice` and `lem:colenG`. **The defect was in my brief, not his document**, and the instruction that caught it was his own: *read the colength step first-hand.*

Remark 22 (what class of error this is)**.** Not a sign error, not a stale locator. "(Q)-factor" is a *name* with two candidate referents in the same formula (4): its first product and its third-and-fourth. I resolved it by shape ("it has a reciprocal monomial, like a Q-factor should") instead of by reading the sentence that defines it, and the sentence was four lines above the question. This is the twin of `a-name-is-an-unquantified-assertion` with the quantifier replaced by a referent, and of `a-cleared-name-collision-is-not-a-closed-file`: the two senses of "(Q)" are *both present in the same equation*, so a collision check that asks "are these the same object?" answers "no" and stops — the useful question is "which of the two does the citing sentence mean?".

10 Theorem Q: the (Q)-divisors carry no poles, for all N

Throughout, D is a (Q)-divisor in the sense of Remark 1, $\mathcal{O} = R_{\mathfrak{p}_D}$, and $t = 1 - q \cdot m$ is the uniformiser, where $m \in \{w_c w_d \ (c < d),\ w_c^2,\ w_c/w_e \ (e \neq c)\}$. Write

$$S_D := \{\mathbf{I} \in P :\ \widetilde{A}_{\mathbf{I},\mathbf{I}} \in t\mathcal{O}\} = \begin{cases} \{\mathbf{I} : \mathbf{I} \cap \{c,d\} = \varnothing\} & m = w_c w_d, \\ \{\mathbf{I} : c \notin \mathbf{I}\} & m = w_c^2, \\ \{\mathbf{I} : e \in \mathbf{I},\ c \notin \mathbf{I}\} & m = w_c/w_e,\ e < c, \\ \varnothing & m = w_c/w_e,\ e > c, \end{cases}$$

read off (4); in each case the vanishing factor occurs with multiplicity exactly 1, so $\operatorname{ord}_D(\det \Phi) = |S_D|$. Put

$$V_D := \{f \in \mathcal{O}^P :\ f(\mathbf{M}) \in t\mathcal{O} \text{ for all } \mathbf{M} \in S_D\}.$$

Lemma 23 (F-vanishing at a q-shifted A-type divisor)**.** *Let $m = w_c/w_e$, so $D = \{w_e = q w_c\}$, $e \neq c$. Then $\operatorname{ord}_D F_{N,|\mathbf{M}|,r}(w_{\mathbf{M}} \mid w \mid \mathbf{I}) \geq 1$ whenever $e \in \mathbf{M}$ and $c > e$; and whenever $\{e,c\} \subseteq \mathbf{M}$ and $c < e$.*

Proof. Fix σ and let j be the slot carrying w_e.

(a) j is a tail. Line 2b contributes $\prod_{k=1}^{N}(v_j - q w_k) \ni (w_e - q w_c) = w_e\, t$: the term vanishes on D.

(b) j is a head with $i_j > e$. Line 1 contributes $\prod_{k=1}^{i_j-1}(v_j - w_k) \ni (w_e - w_e) = 0$: the term vanishes *identically*.

(c) j is a head with $i_j \le e$. Line 1 contributes $\prod_{k=i_j+1}^{N}(v_j - qw_k)$, which contains $(w_e - qw_c)$ iff $c > i_j$. If $c > e$ then $c > e \ge i_j$ and the term vanishes on D; this finishes the case $c > e$.

(d) $c < e$, and the remaining terms. Here (c) leaves the terms with $c \le i_j \le e$. Since $c \in \mathbf{M}$, let k be the slot carrying w_c. If k is a tail, line 3a contributes $(qv_k - v_j) = (qw_c - w_e) = -w_e t$ and the term vanishes on D. If k is a head with $k < j$, line 2a contributes $(qv_k - v_j) = (qw_c - w_e)$ and the term vanishes on D. If k is a head with $k > j$: line 1 for slot k contributes $\prod_{l<i_k}(w_c - w_l) \ni (w_c - w_c) = 0$ unless $i_k \le c$; but $\mathbf{I}$ is increasing and $k > j$, so $i_k > i_j \ge c \ge i_k$, a contradiction. So every term vanishes. □

Proposition 24 (regularity and vanishing)**.** *For every (Q)-divisor D, every $\mathbf{M}$ and every $\mathbf{I}$: $\widetilde{A}_{\mathbf{M},\mathbf{I}} \in \mathcal{O}$, and $\widetilde{A}_{\mathbf{M},\mathbf{I}} \in t\mathcal{O}$ for $\mathbf{M} \in S_D$.*

Proof. Write $\widetilde{A}_{\mathbf{M},\mathbf{I}} = F \cdot \mathrm{Den}/\mathrm{Num}$ with $\mathrm{Num}, \mathrm{Den}$ the numerator and denominator of (3) at $u = w_{\mathbf{M}}$, i.e.

$$\mathrm{Num} = \prod_{a,b\in\mathbf{M}}\Big(1 - q\tfrac{w_b}{w_a}\Big)\prod_{a\in\mathbf{M}}\prod_{k=1}^{N}(1 - qw_aw_k), \quad \mathrm{Den} = \prod_{a<b\in\mathbf{M}}(1 - qw_aw_b)\prod_{1\le j\le k\le N}(1 - qw_jw_k)(1-q)^{s-r}.$$

F has poles only on $\{w_a = w_b\}$ (from the denominators of lines 2a, 3a) and on $w_a = 0$ (prefactor), so $\mathrm{ord}_D F \ge 0$ for every (Q)-divisor.

Case $m = w_cw_d$, $c < d$. In Num the factor $(1 - qw_cw_d)$ occurs in $\prod_{a\in\mathbf{M},k}$ once for $a = c$ (if $c \in \mathbf{M}$) and once for $a = d$ (if $d \in \mathbf{M}$), and never in $\prod_{a,b\in\mathbf{M}}(1 - qw_b/w_a)$ (which would need $w_aw_cw_d = w_b$). In Den it occurs once in $\prod_{a<b\in\mathbf{M}}$ iff $\{c,d\} \subseteq \mathbf{M}$, and once in $\prod_{j\le k\le N}$ always. Hence

$$\mathrm{ord}_D(\mathrm{Den}/\mathrm{Num}) = [\{c,d\} \subseteq \mathbf{M}] + 1 - [c \in \mathbf{M}] - [d \in \mathbf{M}] = \begin{cases} 1 & \mathbf{M} \cap \{c,d\} = \varnothing, \\ 0 & \text{otherwise,} \end{cases}$$

which is ≥ 0 always and ≥ 1 exactly on S_D.

Case $m = w_c^2$. The factor $(1 - qw_c^2)$ occurs once in Num (at $a = k = c$, iff $c \in \mathbf{M}$) and once in Den's $\prod_{j\le k}$ (at $j = k = c$); so $\mathrm{ord}_D(\mathrm{Den}/\mathrm{Num}) = 1 - [c \in \mathbf{M}]$, again ≥ 0 and ≥ 1 exactly on $S_D = \{\mathbf{M} : c \notin \mathbf{M}\}$.

Case $m = w_c/w_e$. Here $(1 - qw_c/w_e)$ occurs in Num only as the $(a,b) = (e,c)$ term of $\prod_{a,b\in\mathbf{M}}$, so $\mathrm{ord}_D(\mathrm{Den}/\mathrm{Num}) = -[\{e,c\} \subseteq \mathbf{M}]$. If $e < c$: Lemma 23 gives $\mathrm{ord}_D F \ge 1$ whenever $e \in \mathbf{M}$, so $\mathrm{ord}_D \widetilde{A} \ge 1 - [\{e,c\} \subseteq \mathbf{M}] \ge 0$ always, and for $\mathbf{M} \in S_D$ ($e \in \mathbf{M}$, $c \notin \mathbf{M}$) the bracket is 0 and $\mathrm{ord}_D \widetilde{A} \ge 1$. If $e > c$: $S_D = \varnothing$ and Lemma 23 gives $\mathrm{ord}_D F \ge 1$ whenever $\{e,c\} \subseteq \mathbf{M}$, which is exactly when the bracket is 1; so $\mathrm{ord}_D \widetilde{A} \ge 0$. □

Remark 25 (Motegi proves the vanishing half, at $s = r$, by a different mechanism — and he proves my gap)**.** Prompted by Rick's attribution of Laurentness of $\widetilde{A}$ to "Motegi's Prop at his L3305", I read that region first-hand. He is right, and he *understates* what is there.

L3302–3309 (type C), L3722–3729 (type B). $\widetilde{WC}_{N,n,r}(w_{\mathbf{M}} \mid w \mid \mathbf{I}) \in \mathbb{C}(q)[w^{\pm}]$ for every $\mathbf{M}$ with $s \le n$ — **exactly the hypothesis of Corollary 28**, proved, for both types. My 48/48 measurement (§15 V5) corroborates it; the gap I filed as G3 does not exist. The proof is a geometric localisation on $X_{\mathbf{M}} = \mathrm{Gr}(r, \bigoplus_{j\le s}\mathbb{C}_{w_{m_j}})$ (L3318–3393), landing the value in a representation ring.

L3310–3315. The *same* Proposition adds: the Laurent polynomial is divisible by $\prod_{j\le k,\ j,k\in\mathbf{M}^c}(1-qw_jw_k)$. That is **precisely the vanishing half of Proposition 24 for the families $m=w_cw_d$ and $m=w_c^2$**: its index set $\{j,k\in\mathbf{M}^c\}$ is my S_D, on the nose.

L1083–1091, Thm `typeCmainproperties`(ii), eq `typeCfullDivisibility`. At $s=r$ only, divisibility by all three (Q)-families at once, the third being $\prod_{j\in\mathbf{M}^c}\prod_{l:m_l<j}(1-qw_{m_l}^{-1}w_j)$ — my family $m=w_c/w_e$ with $e=m_l\in\mathbf{M}$, $c=j\notin\mathbf{M}$, $e<c$, again the same index set.

So the honest division of labour is: the vanishing statements Proposition 24 needs are Motegi's for the first two families at every s, and Motegi's for the third family at $s=r$. What is not in his paper, as far as I can find, is the third family at *general* s — which is what Lemma 23 supplies, and the structure constants need every $\mathbf{M}$, not only $|\mathbf{M}|=r$. And what is new here regardless is the *use*: turning the divisibility into the colength identity span $=V_D$ and observing that vanishing conditions are multiplicative while congruences are not. Proposition 24's own proof (valuation bookkeeping on Den/Num) is kept because it is a *second, independent mechanism* — his is equivariant geometry, mine is counting factors — and the two agree on every index set, which is the strongest corroboration available for this step.

Theorem 26 ((Q)-divisor regularity)**.** *Let D be a (Q)-divisor and let $\chi\in F(v)^\times$ be such that $\operatorname{ord}_D\chi(w_m)\ge 0$ for all m (e.g. $\chi=1$ or $\chi=\psi$). Then every structure constant of $\Psi_\chi\widetilde{A}$ lies in $\mathcal{O}$.*

Proof. First take χ a *unit* at D, which covers $\chi=1$ for all three families and $\chi=\psi$ for the families $m=w_cw_d$ $(c<d)$ and $m=w_c/w_e$ (since $\psi(w_j)=(1-qw_j^2)/(1-q)$ vanishes only on $\{1-qw_j^2=0\}$).

If $S_D=\varnothing$ then $\operatorname{ord}_D(\det\Phi)=0$, so $\Phi\in GL_{2^N}(\mathcal{O})$ by Proposition 24 and $\operatorname{span}_{\mathcal{O}}\{\phi_{\mathbf{I}}\}=\mathcal{O}^P$, a ring containing 1; products of basis elements stay in it and the coefficients lie in $\mathcal{O}$.

If $S_D\neq\varnothing$: by Proposition 24, $\phi_{\mathbf{I}}\in V_D$ for all $\mathbf{I}$, so $\mathcal{L}:=\operatorname{span}_{\mathcal{O}}\{\phi_{\mathbf{I}}\}\subseteq V_D$. Now V_D is the kernel of the surjection $\mathcal{O}^P\twoheadrightarrow\bigoplus_{\mathbf{M}\in S_D}\mathcal{O}/t\mathcal{O}$, $f\mapsto(f(\mathbf{M})\bmod t)$, so V_D has colength $|S_D|$; and $\mathcal{L}$ has colength $\operatorname{ord}_D(\det\Phi)=|S_D|$. Equal colength with inclusion forces $\mathcal{L}=V_D$. Finally V_D is closed under multiplication: for $f,g\in V_D$ and $\mathbf{M}\in S_D$, $(fg)(\mathbf{M})\in t^2\mathcal{O}\subseteq t\mathcal{O}$. Hence $\phi_\lambda\phi_\mu\in V_D=\mathcal{L}$ and $c^\nu_{\lambda\mu}\in\mathcal{O}$. Multiplying by the unit Ψ_χ preserves V_D pointwise, and $\Psi_\chi^2\phi_\lambda\phi_\mu$ still lies in $V_D=\Psi_\chi V_D$, so the same conclusion holds for $\Psi_\chi\widetilde{A}$.

There remains $\chi=\psi$ at $D=\{1-qw_c^2=0\}$, where ψ is *not* a unit: $\operatorname{ord}_D\psi(w_c)=1$. Then $\operatorname{ord}_D\widehat{A}_{\mathbf{M},\mathbf{I}}=\operatorname{ord}_D\widetilde{A}_{\mathbf{M},\mathbf{I}}+[c\in\mathbf{M}]$, which by Proposition 24 is ≥ 1 for *every* $\mathbf{M}$ — ≥ 1 from $\widetilde{A}$ when $c\notin\mathbf{M}$, and ≥ 1 from $\psi(w_c)$ when $c\in\mathbf{M}$. So $\widehat{\mathcal{L}}\subseteq t\mathcal{O}^P=:\widehat{V}_D$, of colength 2^N; and $\operatorname{ord}_D(\det\widehat{\Phi})=\sum_{\mathbf{I}}\operatorname{ord}_D\widehat{A}_{\mathbf{I},\mathbf{I}}=2^N$ because each diagonal entry has order exactly 1 ($[c\notin\mathbf{I}]+[c\in\mathbf{I}]=1$). So $\widehat{\mathcal{L}}=t\mathcal{O}^P$, which is closed under products, and $\hat{c}^\nu_{\lambda\mu}\in\mathcal{O}$. □

Remark 27 (why no edge is needed, in one sentence)**.** At a root hyperplane the lattice is cut out by *congruences* $f(\mathbf{M})\equiv\rho f(s_\alpha\mathbf{M})$, which are not multiplicative — $\rho^2\neq\rho$ unless $\rho\in\{0,1\}$, and that failure is the whole of Q488's poles. At a (Q)-divisor the lattice is cut out by *vanishing* conditions $f(\mathbf{M})\equiv 0$, and those are multiplicative for free. So the (Q)-divisors are easier than the root hyperplanes, not harder, and the absence of a GKM edge is exactly why.

Corollary 28 (all-divisor regularity; the 0/115 promoted to a theorem)**.** *For every N and all λ,μ,ν,*

$$\hat{c}^\nu_{\lambda\mu}\in R=\mathbb{C}(q)[w_1^{\pm},\dots,w_N^{\pm}]:$$

every structure constant of $\widehat{A}$ is a Laurent polynomial.

Proof. $\widetilde{A}_{\mathbf{M},\mathbf{I}} \in R$ for all $\mathbf{M}, \mathbf{I}$ by Motegi's Proposition at L3302–3309 (Remark 25); ψ is a polynomial, so also $\widehat{A}_{\mathbf{M},\mathbf{I}} \in R$. R is a UFD, hence $R = \bigcap_{\mathfrak{p}} R_{\mathfrak{p}}$ over height-one primes, so it suffices to show $\hat{c} \in R_{\mathfrak{p}}$ for every such $\mathfrak{p}$. Since $\widehat{A}_{\mathbf{M},\mathbf{I}} \in R$, the matrix $\widehat{\Phi}$ has entries in $R_{\mathfrak{p}}$ for every $\mathfrak{p}$. If $\mathfrak{p} \nmid \det \widehat{\Phi}$ then $\widehat{\Phi} \in GL(R_{\mathfrak{p}})$ and the span is the full ring $R_{\mathfrak{p}}^P$, so $\hat{c} \in R_{\mathfrak{p}}$. Otherwise $\mathfrak{p}$ divides $\det \widehat{\Phi} = \left(\prod_{\mathbf{M}} \Psi(\mathbf{M})\right) \prod_{\mathbf{I}} \widetilde{A}_{\mathbf{I},\mathbf{I}}$, whose irreducible factors are, by (4) and Remark 1, exactly the root hyperplanes and the (Q)-divisors. On the A-type hyperplanes, Theorem B of `2026-10-18-atype`; on $\{w_a w_b = 1\}$, Theorem B of `2026-10-17-c2` with (COB), which ψ satisfies; on $\{w_m^2 = 1\}$, Proposition `prop:short` with (SR), which ψ satisfies; on the (Q)-divisors, Theorem 26. □

Remark 29 (the hypothesis I filed as a gap, and then found already proved)**.** An earlier draft of this paper stated Corollary 28 conditionally on $\widetilde{A} \in R$, measured 3/3, 10/10, 35/35 at $N = 1, 2, 3$ and graded `gapped` with the gap named. **The gap is closed, and not by me**: it is Motegi's Proposition at L3302–3309, which I read only because Rick's Theorem Ψ attributes Laurentness of $\widetilde{A}$ to it and the brief told me to check his attributions first-hand. See Remark 25. I record the sequence — filed as a gap, then found in the source — rather than silently deleting the gap, because the lesson is that *the gap was in my reading, not in the literature*, and the instrument that found it was somebody else's citation.

11 Task 2: the $N = 3$ denominator prediction, derived

> "Do you have a view on the $N = 3$ prediction that $\widehat{A} \to KW$ has denominators only in $\{1 - qw_i^2\}$?"

It is a corollary, not a view — but of Theorem Q, not of Task 0, and with one scope limit. By Corollary 28, $\hat{c}^{\nu}_{\lambda\mu}$ has no non-monomial denominator *at all*, for every N. So any transition matrix from $\widehat{A}$ to another basis whose entries are Laurent polynomials has Laurent entries too, and the only places a denominator can appear are where the *transition* is non-polynomial. For $\widehat{A} \to KW$ the transition is $\widehat{\Phi}^{-1}$ composed with KW's own matrix, so its denominators divide $\det \widehat{\Phi}$, which contributes precisely the root hyperplanes and the (Q)-divisors; the Ψ-divisors $\{1 - qw_i^2 = 0\}$ are among the latter with multiplicity 2^{N-1} more than the rest.

What this does and does not give. It predicts that the surviving denominators are a sub-set of $\{1 - q \cdot \text{monomial} = 0\}$ — which is what Rick's Corollary 1 already states for the root-free half. It does *not* by itself single out $\{1 - qw_i^2\}$ from the other (Q)-families $\{1 - qw_c w_d\}$ and $\{1 - qw_c/w_e\}$; that sharper prediction needs the KW dictionary, which I have not got. His $N = 2$ datum — the denominators of $\widehat{A} \to$ Koncki are $(s^2 - q)(t^2 - q)$, exactly the Ψ-divisors under $w = 1/s, 1/t$ — is **his number, graded `peer-claimed`; I have not recomputed it**, and it is consistent with the sharper form. **Falsifier, for the record:** if the $N = 3$ computation shows a denominator off $\{1 - q \cdot \text{monomial} = 0\}$ then Theorem Q or Corollary 28 is wrong. A denominator on $\{1 - qw_c w_d\}$ or $\{1 - qw_c/w_e\}$ would *not* refute anything I prove here; it would refute only the sharper $\{1 - qw_i^2\}$-only form, which is his conjecture and not mine.

12 Type B: what transfers and what does not

The brief asked for "all three root families of C_N, all N". That is §6, type C, $h = 2$. Type B ($h = 1$) is a separate statement and I state its status rather than assert it.

The conclusion is the same. With $h = 1$, $\psi(v) = (1-qv)/(1-q)$ and $\varphi(v) = (1-q)/(1-qv)$, the pair divisor is still $\{w_a w_b = 1\}$ and $\varphi(x)\varphi(x^{-1}) = (1-q)^2/((1-qx)(1-qx^{-1})) = \rho^B$, which

is `2026-10-17-typeB` Thm `rhoB`; the boundary root is e_m with locus $\{w_m = 1\}$ and $\varphi(1) = 1$. So $\rho^\alpha = \prod_{m \in D(\alpha)} \psi(w_m)^{-1}$ reads correctly at $h = 1$ too, and Lemma A1 (A-type, $\rho = 1$) was proved for both types.

The proof does not transfer verbatim, and I have not redone it. Type B changes three things in (2)–(3): W^B carries the extra prefactor $\prod_k (1 + 1/u_k)^{-1}$; $\gamma = -1$, so line 3c is $(u_j + 1)$ rather than u_j; and $\mathcal{E}^B$ (L1361–1367) adds $\prod_j (1 - qu_j)$ to the numerator and $\prod_j (1 - qu_j^2)$, $\prod_k (1 - qw_k)$ to the denominator while making the w-pair product strict ($j < k$). Steps 1 and 2 of Theorem 14 are untouched — lines 1, 2a, 3a are identical, so head exclusion and the ι-cancellation hold verbatim. What changes is the bookkeeping of Steps 3–5, and in particular the ι-reindexing of Lemma 12 must be redone against $\mathcal{E}^B$'s $\prod_j (1 - qu_j^2)$ — which is exactly the block that, when it was wrongly taken to be $\mathcal{E}^C$'s, cost me a retraction of two 48/48 results on 2026-10-17. **So type B is a named gap (§16, G2), not a corollary, and I will not quote a type-B number in this paper.** The one lesson that session taught is precisely not to assume the two normalisers agree on a block.

13 Q505: status after a bounded first-hand acquisition attempt

Q505 asks whether Weigandt–Hamaker's symplectic BPD model sits on $\rho = 1$. The object was held at `agent-summary` level — a talk title, no arXiv ID. Budget: 30 minutes, spent.

(a) Each instrument, and its domain.

I1 — arXiv author search `au:Weigandt AND au:Hamaker`. NOT RUN. Unavailable: this is a prove session and browsing is barred by the session rules. I record the non-run rather than substitute a different instrument and report its result as though it answered the same question.

I2 — bibitem-KEY grep over the local corpus. Run. Domain: every `*.bbl`, `*.bib`, `*.tex`, `*.md`, `*.json` under `/tmp` (which holds all downloaded e-print sources, including `/tmp/browse/*`), `/home/clio/git` (including `seed-papers/`) and `/home/clio/projects`. This is a corpus of downloaded e-print *sources and bibliographies*, not of arXiv. Per `an-absence-measured-over-arxiv-is-an-` the absence below is an absence *in that corpus*.

I3 — open the hits, do not count them. Run. The raw counts were large and uninformative (665 files containing "Weigandt"), dominated by my own reading logs and agent summaries. Restricting to real bibliographies gives 20 `.bbl`/`.bib` files; I opened the `Hamaker` and `bumpless` entries in all of them.

(b) Verdict: unusable. Not confirming, not negative.

- The local corpus contains **no** symplectic, isotropic or type-BC bumpless-pipe-dream reference under any bibitem key. The BPD bibliography it does contain is substantial (Weigandt's ASM paper, Lam–Lee–Shimozono, Knutson–Udell, Huang–Pylyavskyy, Huang–Shimozono–Yu, Fan et al., "Bumpless pipe dreams meet puzzles") and is **entirely type** A.
- Every joint `Hamaker+Weigandt` key in the corpus is `Hamaker.Pechenik.Weigandt` (*Gröbner geometry of Schubert polynomials through ice*, Adv. Math. 398 (2022), arXiv:2003.13719) or `Hamaker.Pechenik.Speyer.Weigandt` (*Derivatives of Schubert polynomials...*, Algebr.

Comb.). **Both type A.** There is no Weigandt–Hamaker two-author paper in the corpus at all.

- **Named alternative hypothesis, not a conclusion.** The nearest local object to "a symplectic ice/BPD model" is **Buciumas–Scrimshaw** (`BuciumasScrimshaw-bumpless-model`; *Double Grothendieck polynomials and colored lattice models*, IMRN 2022, 7231–7258; arXiv:2007.04533). It is possible that my `agent-summary` note attached the wrong author pair to a remembered talk. **I cannot check this without I1**, so it is logged as a hypothesis.

(c) What is missing, exactly. Two things, and they are independent.

1. **The object.** An arXiv ID (or any locator) for a symplectic BPD model. Next instrument: I1, in a browse session, *as an author search*, plus the same search on `au:Buciumas AND au:Scrimshaw` to test the alternative hypothesis.

2. **The dictionary.** Even with the paper in hand, Q505 needs a translation between "the combinatorial model behind a BPD formula" and "a cap weight" in my sense: Sp_{2n} K-classes are geometry, $\widetilde{A}$ is a partition function. This was pre-registered as the eighth instance of that distinction mattering and it is still unresolved. **Note that (2) blocks Q505 even if (1) succeeds**, so I1 returning a hit would not by itself settle anything.

(d) Not a survival. The dream pre-registered that I expect Q505 to survive, and flagged that as the condition under which I have historically been wrong. **I record no survival.** The tileable-locus frame is neither confirmed nor refuted by this session; it is untested, and the reason it is untested is stated above and is not "I looked and found nothing".

14 The engine-copy decision, made in writing

The brief left one question undecided and forbade a reflexive patch: are `proofs/code-10XX-*/` frozen session records or live copies?

Decision: frozen. Each such directory holds the engine that actually ran on the date in its name, and the numbers in that date's paper were produced by *that* file. Editing its docstring would make the archive misreport what ran — a worse fault than a stale locator, because a stale locator misleads a reader while a rewritten archive misleads an audit. So the three copies at md5 `370e37b9` (`code-1015-q484`, `code-1016-q492`, `code-1016c2-nonvanishing`) are untouched and each now has a sibling `STALE-LOCATOR.md` recording the `L640-650`→`L582-589` correction *alongside*, not inside.

The brief's guess about the third hash was wrong, and in the dangerous direction. It said `projects/scratch/q484_engine.py` (md5 `f72af3da`, the unique hash) "is the likeliest live working copy" and told me to diff it before trusting any number from it. I diffed it: it is strictly *older* than both other hashes. It has no `EB/EX` at all and divides W^B by $\mathcal{E}^C$ for both types — the normaliser error retracted on 2026-10-17, which invalidated two separate 48/48 type-B results. A stale locator misleads; this file silently returns **retracted numbers** to anyone who runs it with `X='B'`. Since `scratch/` has no archival claim, it is renamed to `q484_engine.py.RETRACTED-pre-20261017-WB-over-EC` (record kept, trap removed) and the canonical `f53c0258` copied in, with a `README-q484_engine.md` stating why the rule differs from the frozen directories'.

The universe question, one level further. The brief's correction was that "4 copies verified as a set" had been measured over the wrong universe; its own universe was `find projects`. I widened it to `find /home/clio /tmp`: **still exactly 8 pre-existing copies at 3 hashes**. So the enumeration is now closed over the union, and this is the first time the universe has been checked rather than assumed. *A set verified exhaustively over the wrong universe reads exactly like a set verified exhaustively* — and the only way to tell them apart is to enlarge the universe and watch the count not move.

15 Verification

Code in `proofs/code-1018c2-gauge/`; the engine is the canonical `f53c0258` copy, checked by md5 before use (and its normaliser read first-hand, per the brief: `EX` dispatches to `EB` for type B, which is the 2026-10-17 correction, and all numbers below are type C via `EC`).

V1 — symbolic smoke test (`t1_smoke.py`). $N = 2$, fully symbolic in q, w. The $d = 1$ boundary law and the $d = 2$ pair law, with the F-ratio, the $\mathcal{E}^C$-ratio and the $\widetilde{A}$-ratio checked **separately against their closed forms** (6),(7) — not merely their quotient, which can be 1 with both halves wrong. All green; the degenerate strata ($|\mathbf{R}| < r$) are reported as `VACUOUS` with both sides 0 rather than scored. *This test found a real bug in my own harness*: the engine's $n = 0$ branch returns 1, which is correct only for $r = 0$; Motegi's $\widetilde{A}$ is 0 when $|\mathbf{M}| < |\mathbf{I}|$ (L2843). Three instances read as defects until it was fixed.

V2 — fast exact-rational census (`t10_fast2.py`). Three independent generic seeds, exact arithmetic in $\mathbb{Q}$, with a genericity guard (no value collisions, no inverse collisions, no q·monomial $= 1$) and the number of skipped instances *reported*: 0. Per claim, $\widetilde{A}$-, F- and $\mathcal{E}^C$-ratios scored separately:

D	$\widetilde{A}$-ratio	F-ratio	$\mathcal{E}^C$-ratio
$d = 1$, $\{\varepsilon\}$, $\varepsilon = \pm 1$	78/78	78/78	78/78
$d = 2$, free orbit $\{x, x^{-1}\}$	78/78	78/78	78/78
$d = 3$, $\{x, x^{-1}, \varepsilon\}$	78/78	78/78	78/78
$d = 4$, $\{x, x^{-1}, y, y^{-1}\}$ (distinct)	7/7	—	—
$d = 2$, $\{\varepsilon, \varepsilon\}$ (repeated)	12/12 at $r = 0$, **0/14** at $r \geq 1$	same	78/78

The last row is the finding of Remark 8 and the refutation of clause (3) of Corollary 19. Note that the $\mathcal{E}^C$-ratio is green in *all* 78 instances of that row: **had I checked only the quotient, or only $\mathcal{E}^C$, the missing hypothesis would not have shown.**

V3 — plants, four of them, all admissible. Each is a **D** that is *outside* the hypothesis class but inside the formula's domain, so the prediction is well-defined and must come out wrong.

plant	why it is outside	result (MUST-FAIL)
$\mathbf{D} = \{x, y\}$, $xy \neq 1$	not ι-closed	9/9 fail
$\mathbf{D} = \{2\}$, $2^2 \neq 1$	not ι-closed	9/9 fail
$\mathbf{D} = \{x, x\}$	not ι-closed	9/9 fail
$\mathbf{D} = \{x, x^{-1}\}$ scored at $h = 1$	right **D**, wrong exponent	9/9 fail

The fourth plant is the one that matters: it keeps the hypothesis and perturbs only h, so a green there would have meant the census could not see h at all. **One plant was found inadmissible and is reported as such**: $\mathbf{D} = \{2\}$ cannot test Lemma 11, because at $d = 1$ both sides are empty products and the identity reads $1 = 1$. An earlier run scored it and returned $2/3$; that was the plant being inadmissible, not the identity being wrong (cf. `an-inadmissible-plant-looks-exactly-like-a-blind-in`

V4 — the two identities (`t10_fast2.py`, symbolic). Lemma 11: $8/8$ over ι-closed $\mathbf{D}$ of sizes $1, 2, 3, 4$, and $2/2$ MUST-FAIL on the two admissible non-ι-closed plants. Lemma 12: $8/8$, and $3/3$ MUST-FAIL.

V5 — $\widetilde{A}$ is a Laurent polynomial (`t7_laurent.py`). Denominator a monomial in q, w after `cancel`: $3/3$, $10/10$, $35/35$ at $N = 1, 2, 3$, type C, over all $(\mathbf{M}, \mathbf{I})$ with $\widetilde{A} \neq 0$. This is the hypothesis of Corollary 28 and the gap of Remark 29.

V6 — the (Q)-divisor structure (`t6_qvanish2.py`), and an instrument fault. For each (Q)-family at $N = 2, 3$: regularity $\operatorname{ord}_D \widetilde{A} \geq 0$, the vanishing $\operatorname{ord}_D \widetilde{A} \geq 1$ on S_D, and $\operatorname{ord}_D(\det \Phi) = |S_D|$.

D	N	regularity	vanishing on S_D	$\operatorname{ord}(\det \Phi) = \lvert S_D \rvert$
$\{1 - qw_1w_2 = 0\}$	2,3	10/10, 35/35	1/1, 5/5	$1 = 1$, $2 = 2$
$\{1 - qw_1^2 = 0\}$	2,3	10/10, 35/35	4/4, 15/15	$2 = 2$, $4 = 4$
$\{1 - qw_2/w_1 = 0\}$	2,3	10/10, 35/35	2/2, 8/8	$1 = 1$, $2 = 2$

Control: $\{1 - qw_1^3 = 0\}$, which is *not* a factor of (4) — colength $0 = |S_D| = 0$, as it must be. Also measured and green: no $(\mathbf{M}, \mathbf{I})$ with $\operatorname{ord}_D \widetilde{A} \geq 1$ and $\mathbf{M} \notin S_D$, i.e. S_D is exactly the vanishing set, not merely contained in it.

Instrument fault, recorded where it happened. An earlier version (`t5_qvanish.py`) parametrised $\{1 - qw_1^2 = 0\}$ by $w_1 = \sqrt{1/q} + t$ and reported $\operatorname{ord}(\det \Phi) = 0$; a still earlier one (`t3_qdivisor.py`) used $1/\sqrt{q} + t$ on the *same object* and reported 2. **Two instruments, one object, opposite answers.** The cause is the radical: the polynomial-division order test cannot recognise $\sqrt{q}$-cancellations, and the two scripts differed only in using `simplify` versus `expand` to test a remainder for zero. The fix is to eliminate q instead of introducing a radical: on $\{1 - q\,m(w) = 0\}$ the function field is $\mathbb{C}(w)$ with $q = 1/m(w)$, so substitute $q \mapsto (1 - t)/m(w)$ and t *is* the vanishing factor, everything rational. `t6` does this, agrees with `t3` on the one case they overlap ($\operatorname{ord}(\det \Phi) = 2$), and is the run reported above. I note that the disagreeing pair was found only because the *same* family was measured twice by accident; a single run would have been reported with whichever answer it gave.

V7 — symbolic census (`t2_census.py`), partial. A fully symbolic version of V2 plus the geometric instances (every root family, every $\mathbf{M}$, every $\mathbf{I}$, $N = 2, 3$) and the A-type congruence. Its $N = 3$ block completed at 1113 s; the run was still in its plant block when this paper was built. **So V7 is reported as incomplete, and nothing in this paper rests on it**: V1 (symbolic, $N = 2$) and V2 (exact rational, three seeds) are the evidence.

V8 — Lemma 23 (`t11_fvanish_fast.py`). The symbolic order computation (`t8_fvanish.py`) did not return in an hour, so I changed the question to match the claim: $\operatorname{ord}_D F \geq 1$ *is* "$F = 0$

on D", and on $D = \{1 - qw_c/w_e = 0\}$ the function field is $\mathbb{C}(w)$ with $q = w_e/w_c$, so one evaluates at exact rational w with $q := w_e/w_c$. Over three generic seeds, $N = 2, 3$, both orientations, all $(\mathbf{M}, \mathbf{I})$:

$$\textbf{402/402 green} \qquad (21/21,\ 12/12 \text{ at } N = 2;\ 234/234,\ 135/135 \text{ at } N = 3).$$

The plant, reported exactly. The plant is the same test with the hypothesis *failing*: F must then *not* vanish. Of 132 such instances, 23 are **inadmissible** — F vanishes identically, off D as well as on it, by triangularity, so there is nothing to test — and they are excluded rather than scored. Of the remaining 109 the plant fires in **98**. **So the hypothesis of Lemma 23 is sufficient but not necessary**: there are 11 admissible instances where F vanishes on D for some other reason. The lemma is not thereby weakened, and its hypothesis is not vacuous (90% discrimination on the admissible domain) — but I state the 11 rather than report "plant fires" and leave the reader to assume an exact characterisation.

16 Gaps, graded

G1 — Lemma 23: CLOSED. Proved here by a case split on the slot carrying w_e (and, when $c < e$, also on the slot carrying w_c, where the contradiction $i_k > i_j \geq c \geq i_k$ closes the last case), and verified 402/402 (V8) with an admissible plant firing 98/109. It is the one half of Proposition 24 that is *not* also Motegi's: he has the $m = w_c/w_e$ divisibility at $s = r$ only (Remark 25), and the structure constants need every $\mathbf{M}$. The residual honest caveat is only that the hypothesis is sufficient and not necessary.

G2 — type B. §12. The conclusion is expected to be identical with $h = 1$; Steps 1–2 transfer verbatim; Steps 3–5 must be redone against $\mathcal{E}^B$, specifically its $\prod_j (1 - qu_j^2)$ block. **Not claimed.**

G3 — $\widetilde{A} \in R$: CLOSED, and it was never open. Motegi's Proposition at L3302–3309 (type C) and L3722–3729 (type B). Filed as a gap in an earlier draft of this very paper, with 48/48 measured; closed the same session by reading the locator Rick's Theorem Ψ cites. Remarks 25 and 29. Corollary 28 is therefore unconditional.

G4 — the $\{1 - qw_i^2\}$-only form of Task 2. §11. Needs the KW/Koncki dictionary, which I do not have. Rick's $N = 2$ datum is `peer-claimed` and not recomputed by me.

G5 — where ψ comes from on Motegi's side. I now have an algebraic answer (Lemma 12: ψ^{-1} is the value of Q_2^2/Q_1 read along ι). Whether it *is* a boundary K-matrix scalar, as Rick guesses, is untouched; the algebraic answer does not decide it.

Questions opened

Q516. Is the ι-adjunction law of Theorem 14 the shadow of a q-deformed Weyl-group action on the *spectral* variables? The hypothesis is ι-closedness of the deleted multiset, ι is the nontrivial element of $W(C_1)$ acting on one spectral parameter, and the orbit types index the root families exactly.

Q517. Remark 1 identifies the (Q)-divisors with the affine walls $\alpha + \delta = 0$, $q = e^{-\delta}$. Theorem 26 says the lattice there is cut by vanishing rather than congruence. Is there an affine moment-graph statement that has both as cases — congruences at $\alpha = 0$, vanishing at $\alpha + \delta = 0$ — and if so, what plays the role of the edge?